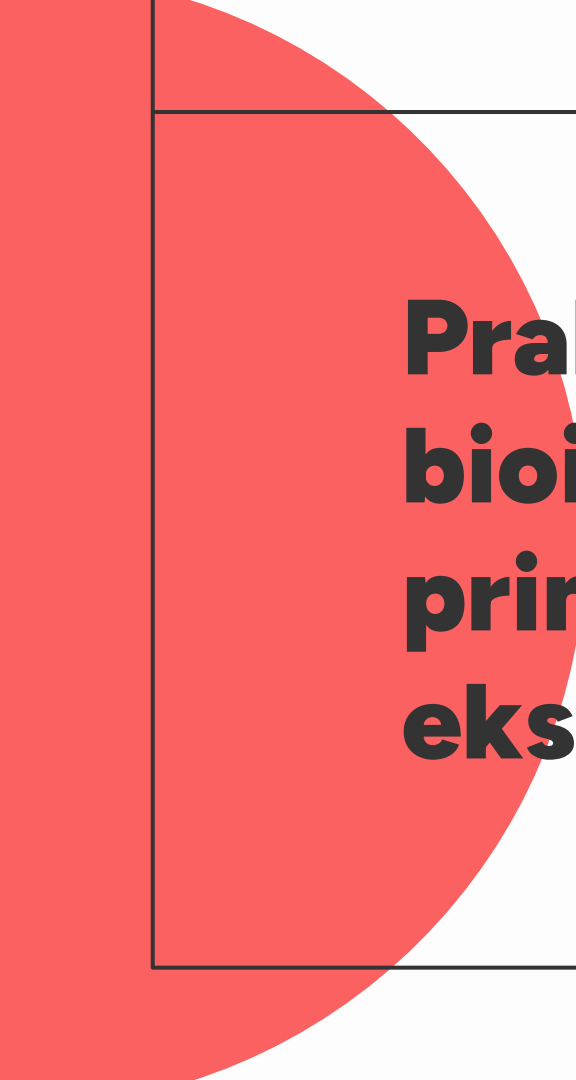




KUZMAN CONSULTING

BIOINFORMATICIANS AT YOUR SERVICE

A large red circle is partially visible on the left side of the slide, overlapping the white content area.

Praktičan uvod u osnove bioinformatike kroz primjer analize genske ekspresije

SILICATE CHEMISTRY IS SECOND NATURE TO US GEOCHEMISTS, SO IT'S EASY TO FORGET THAT THE AVERAGE PERSON PROBABLY ONLY KNOWS THE FORMULAS FOR OLIVINE AND ONE OR TWO FELDSPARS.

OF COURSE. AND QUARTZ, OF COURSE.

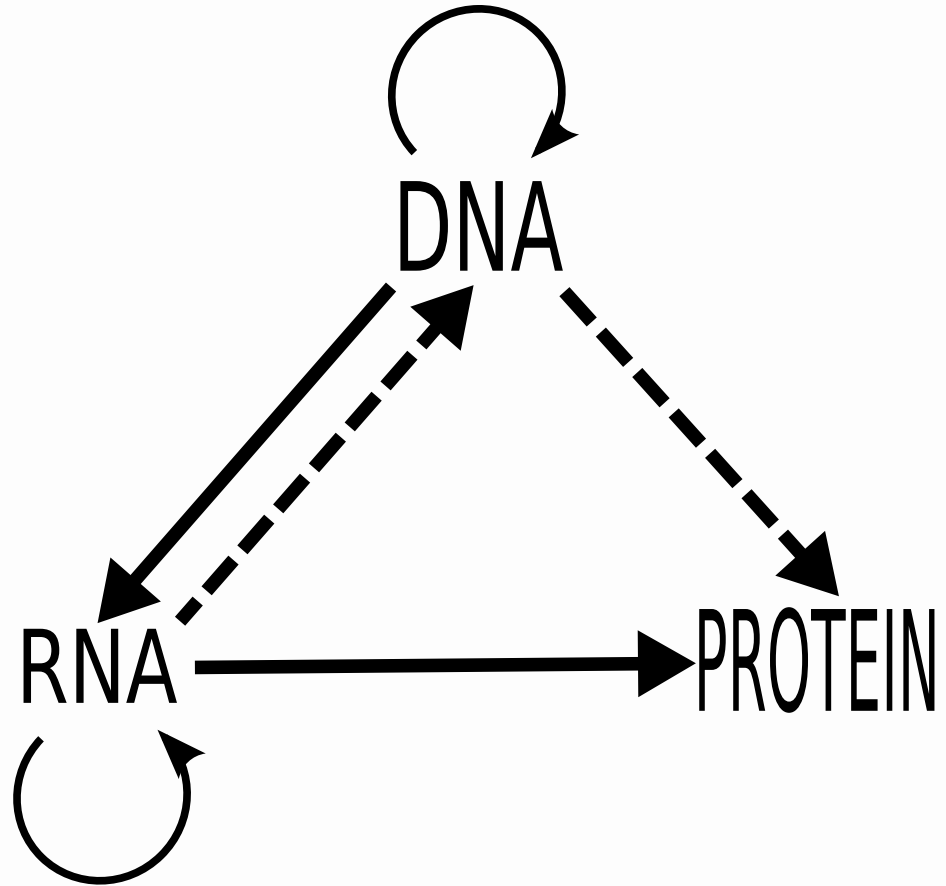


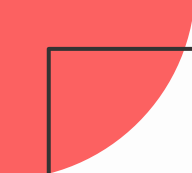
EVEN WHEN THEY'RE TRYING TO COMPENSATE FOR IT, EXPERTS IN ANYTHING WILDLY OVERESTIMATE THE AVERAGE PERSON'S FAMILIARITY WITH THEIR FIELD.

<https://xkcd.com/>

Centralna dogma

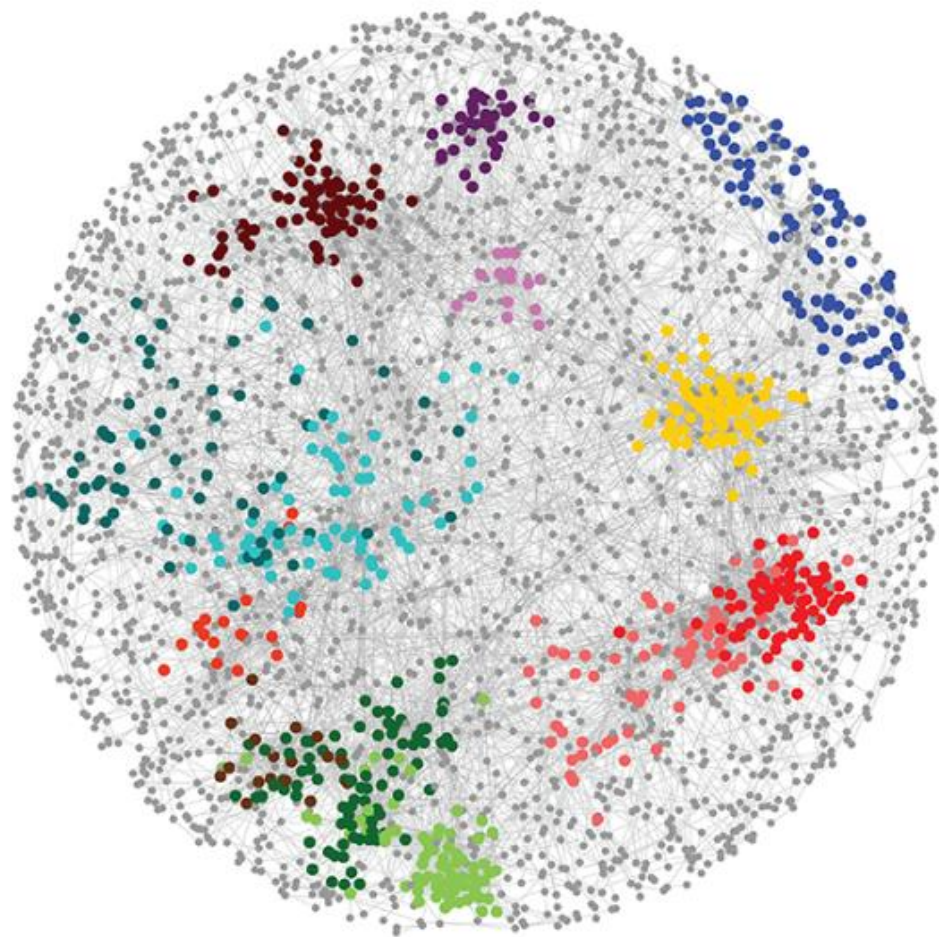
- DNA → RNA → protein
- danas znamo da su stvari složenije od toga...
- Crickova dogma 1958. – smjer prijenosa informacije





Istraživanje u molekularnoj biologiji: klasični pristup

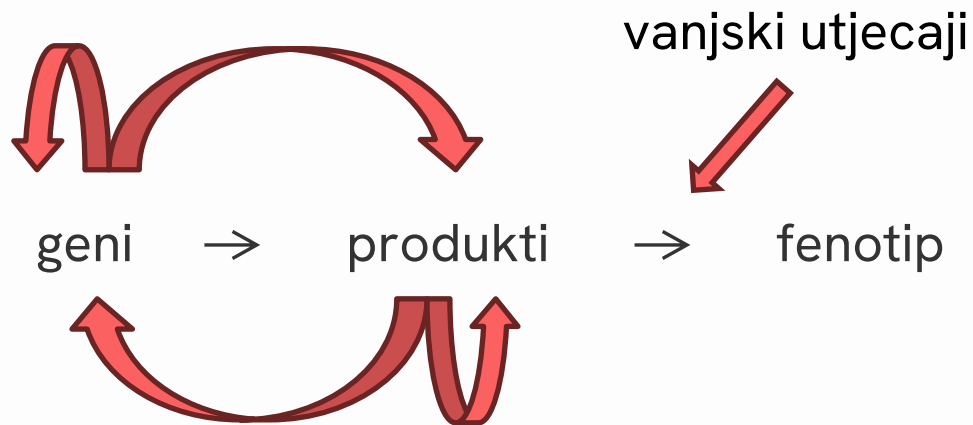
gen → produkt → fenotip



Klasični pristup

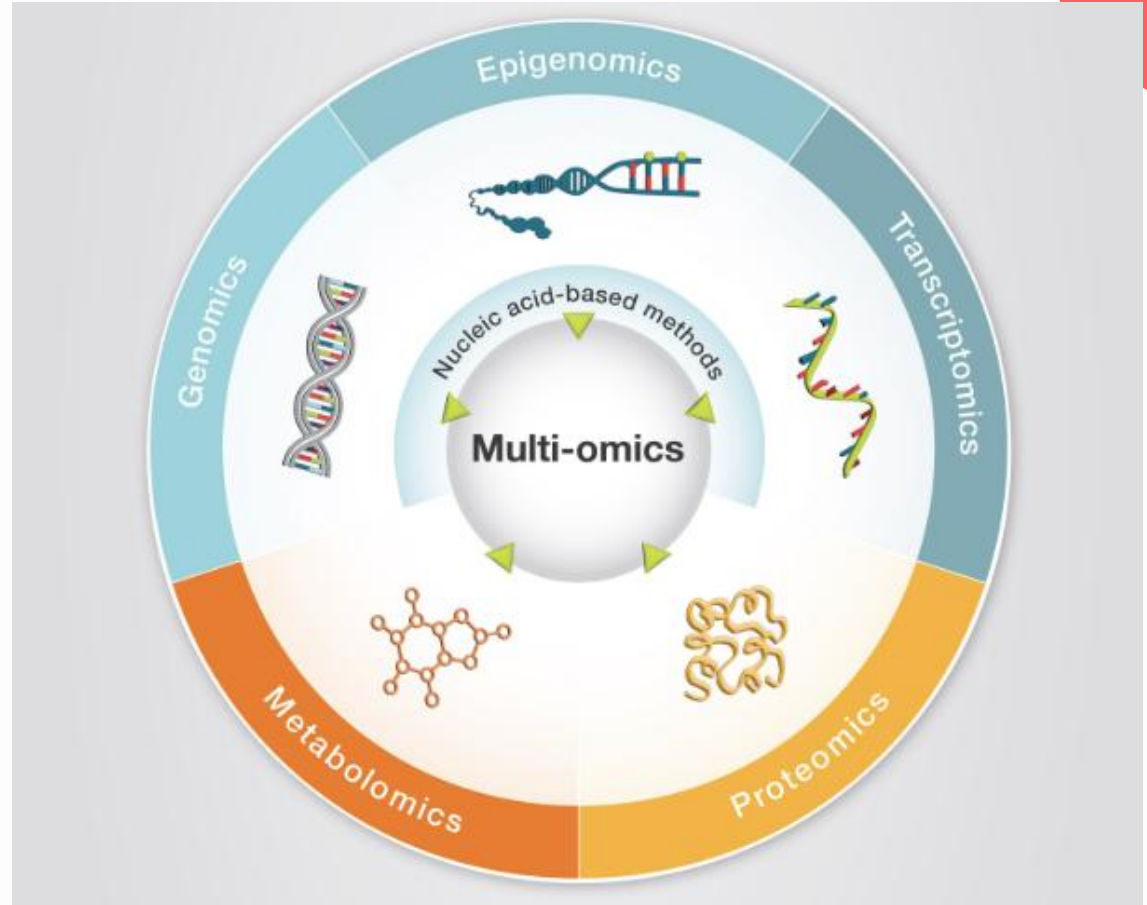
gen $\rightarrow$ produkt $\rightarrow$ fenotip

Moderni pristup



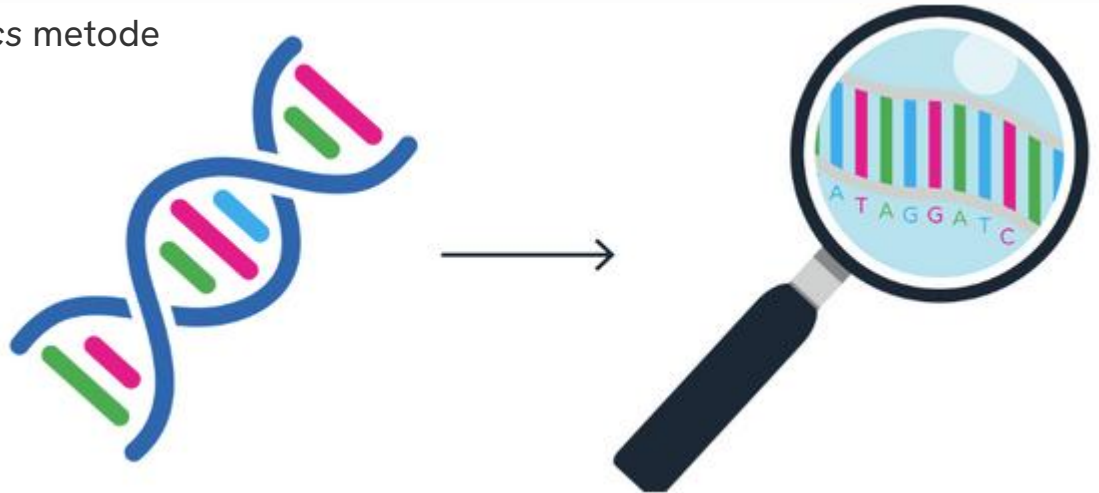
OMICS metode

- genomika
- **transkriptomika**
- proteomika
- metagenomika
- epigenomika
- ...



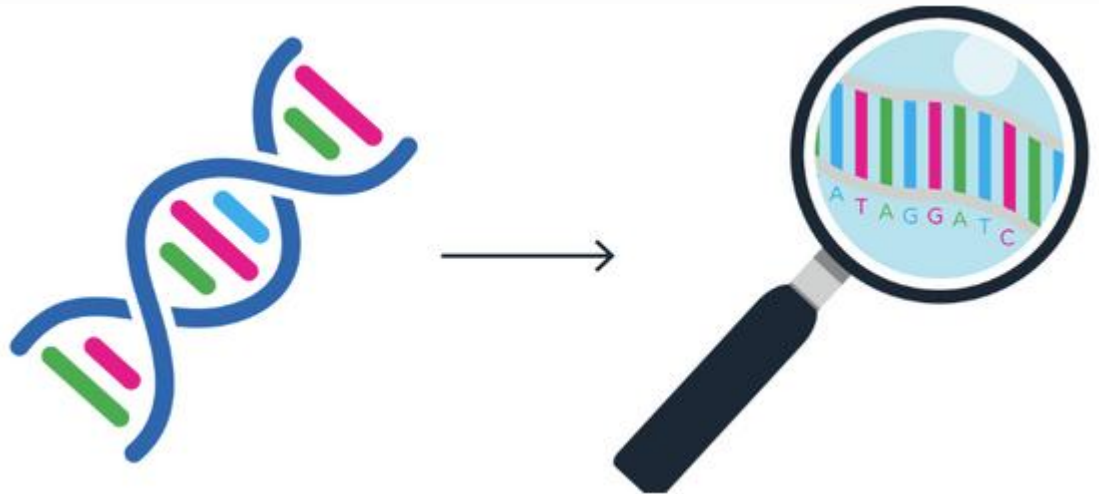
Genomika

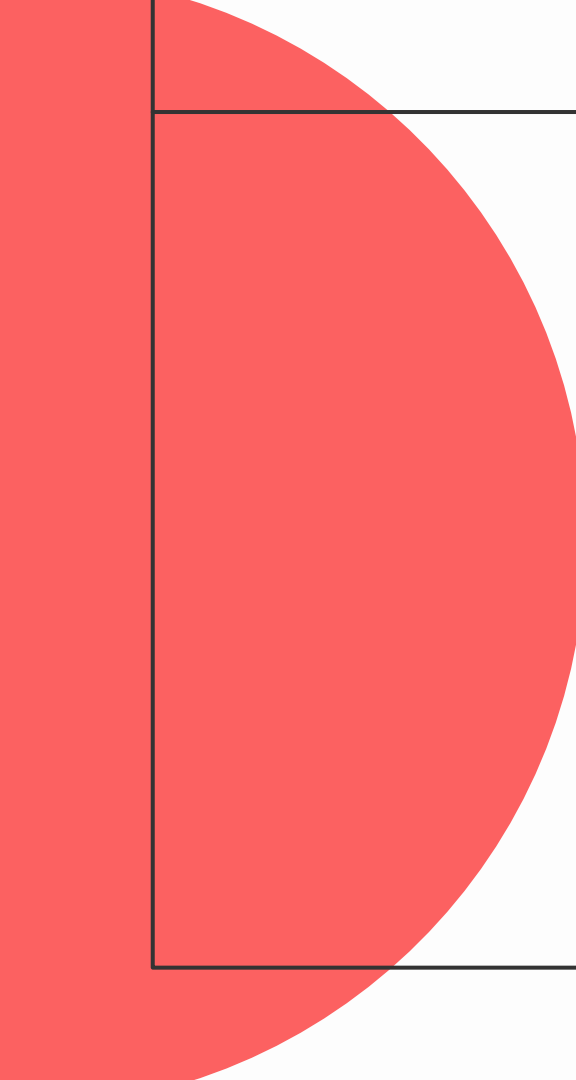
- znanost koja se bavi proučavanjem svih gena / cjelokupnog genetičkog materijala organizma
- struktura i funkcija genoma: kako geni rade, genske interakcije, interakcije s okolišem, evolucija, funkcija i mehanizmi nekodirajućih sekvenci...
- “krovni pojam” i za druge *omics* metode
- široka primjena



Transkriptomika

- proučavanje transkripata prisutnih u organizmu / stanici
- kvalitativni i kvantitativni aspekt
- ovisno o trenutnim uvjetima
- centralna tehnologija: RNA-seq
- *single-cell*



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Sekvenciranje

Sekvenciranje

- proces određivanja slijeda nuklotida u DNA (ili RNA)
- ciljano (targeted) ili *whole-genome*
- podjela:
 1. prema primjeni
 2. prema tehnologiji



Sekvenciranje

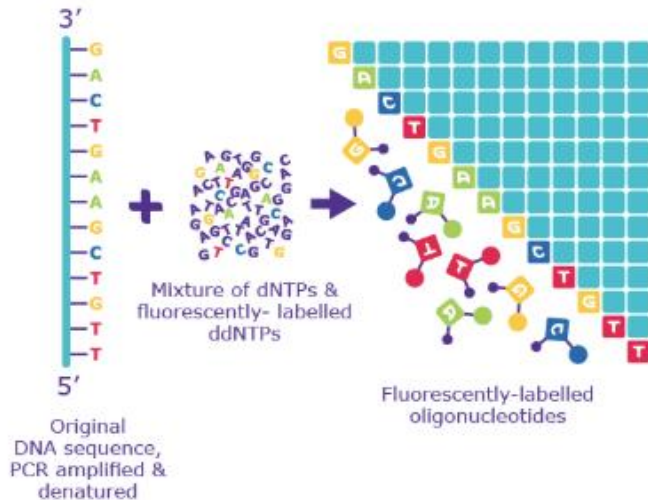
- podjela:
 1. prema primjeni
 2. prema tehnologiji

WGS, WES, RNA-Seq, ChIP-Seq, bisulfite,
RAD-Seq, ATAC-Seq, GRO-Seq, CAGE, CLIP,
single-cell, 3D, 4D, Hi-C, DamID-Seq...

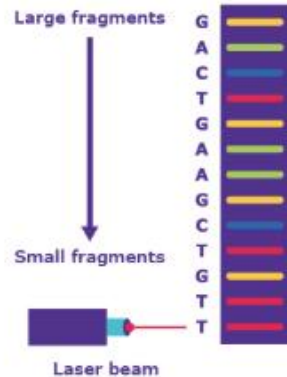


Prva generacija: Sanger

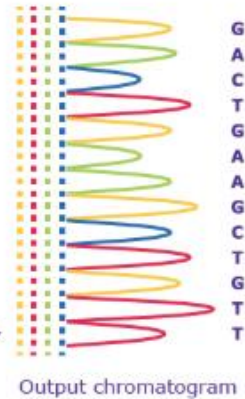
1 PCR with fluorescent, chain-terminating ddNTPs



2 Size separation by capillary gel electrophoresis



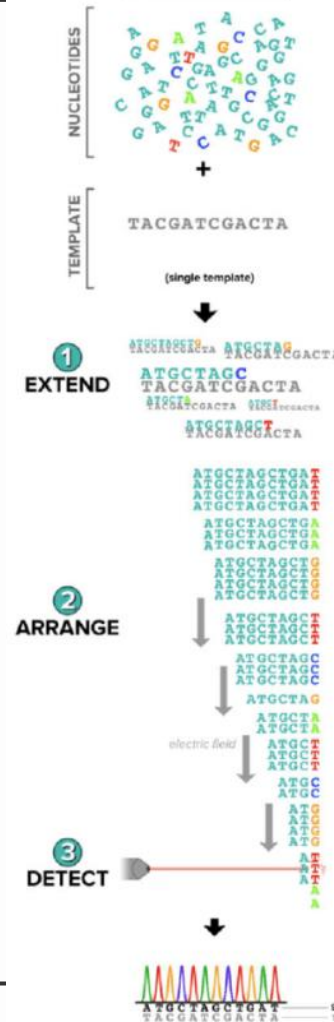
3 Laser excitation & detection by sequencing machine



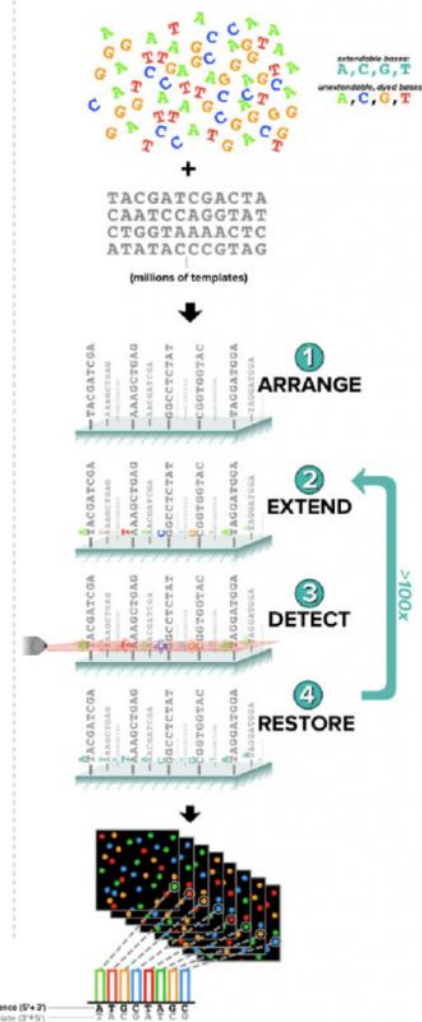
Next generation sequencing

- reverzibilni bloker → nastavak reakcije!
- umnažnje na 2D podlozi → klasteri → više molekula po reakciji!

SANGER:



NGS:



NGS: razvoj

2004. 454/Roche pirosekvenciranje



NGS: razvoj

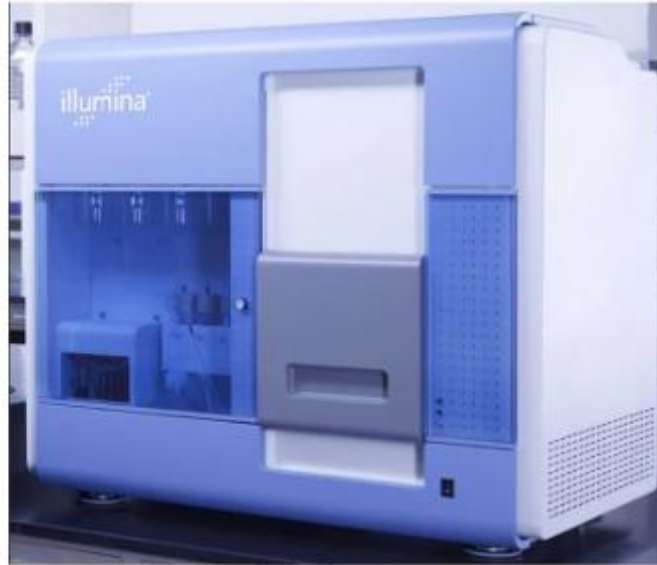
2004. 454/Roche pirosekvenciranje

2006. Solexa Genome Analyzer



NGS: razvoj

2007. Illumina kupuje Solexu



NGS: razvoj

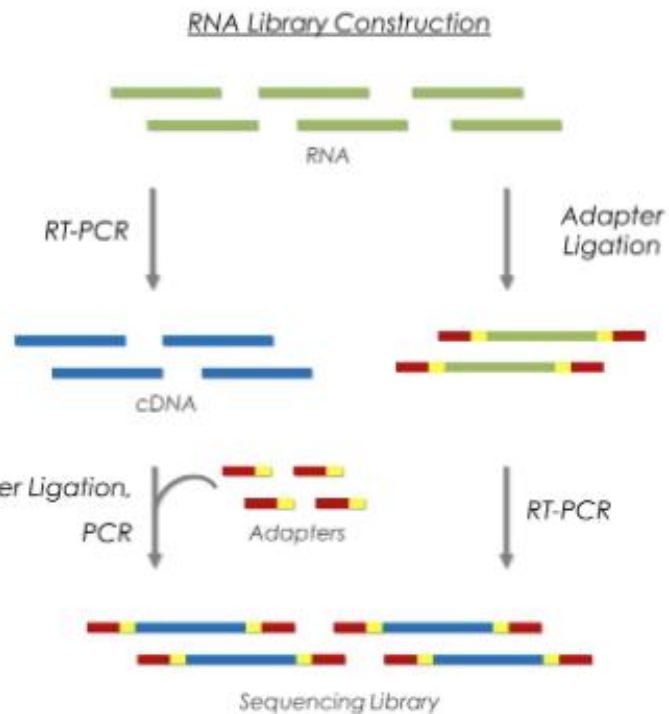
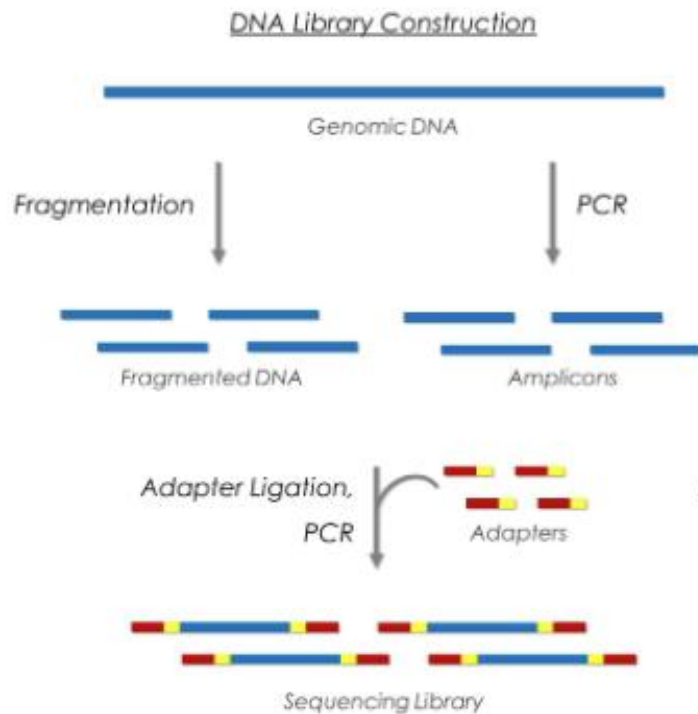
2004. 454/Roche pirosekvenciranje

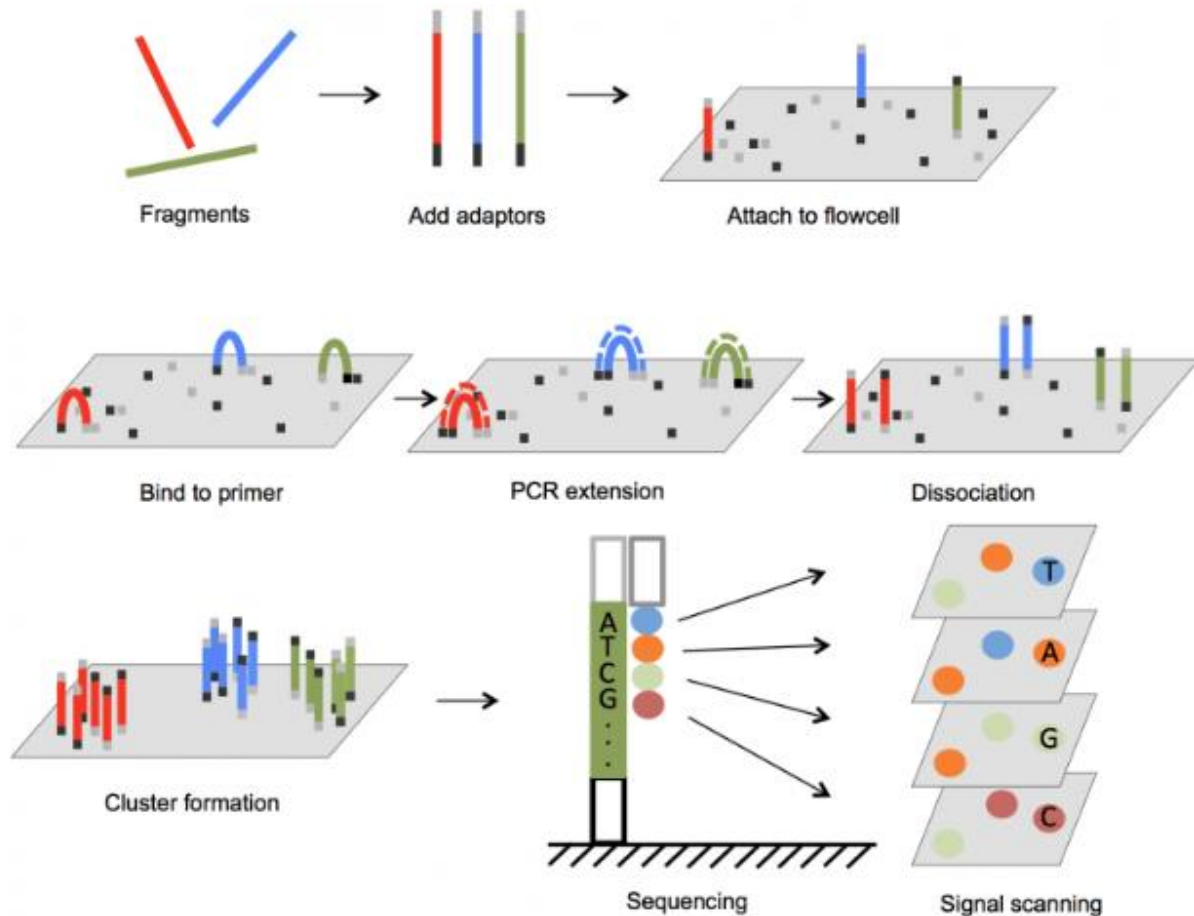
2006. Solexa Genome Analyzer

2014. Illumina HiSeq



Priprema knjižnica za NGS

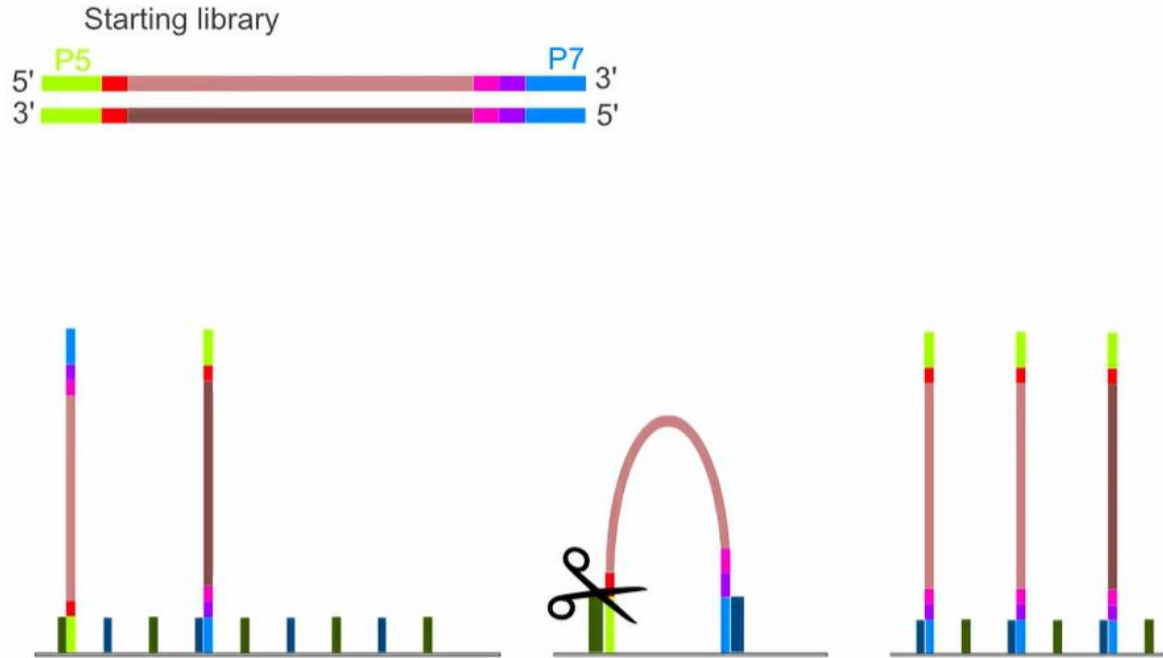




NGS

- *flowcell*
- bridge amplification

Single-end vs paired-end



Single-end vs paired-end

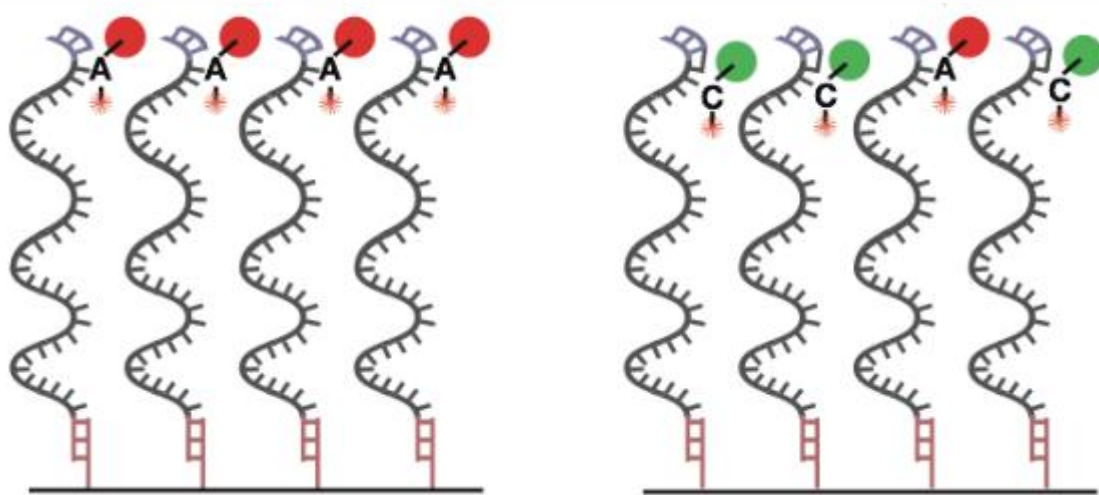
Single-End reads



Paired-Ends reads



Ograničenje NGS-a: duljina *readova*

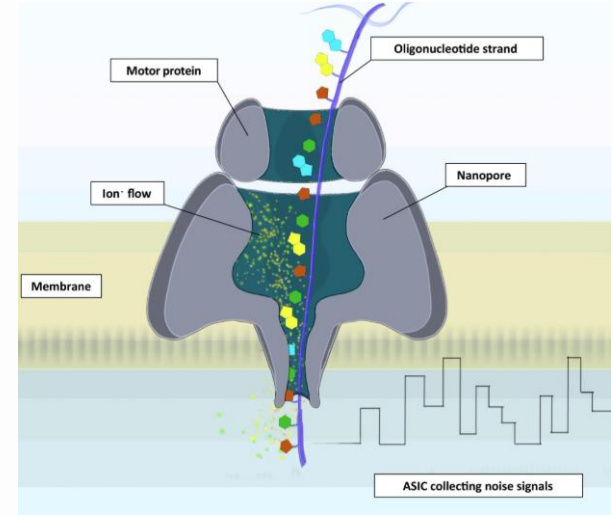


dephasing

Sekvenciranje treće generacije

- dugačka očitavanja – stotine kb
- bez umnažanja
- veća pogreška od Illumine

Oxford
Nanopore



PacBio

